

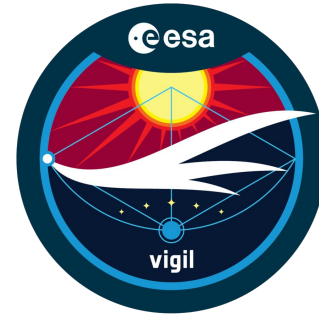
ESA Vigil

Silvia Kostárová

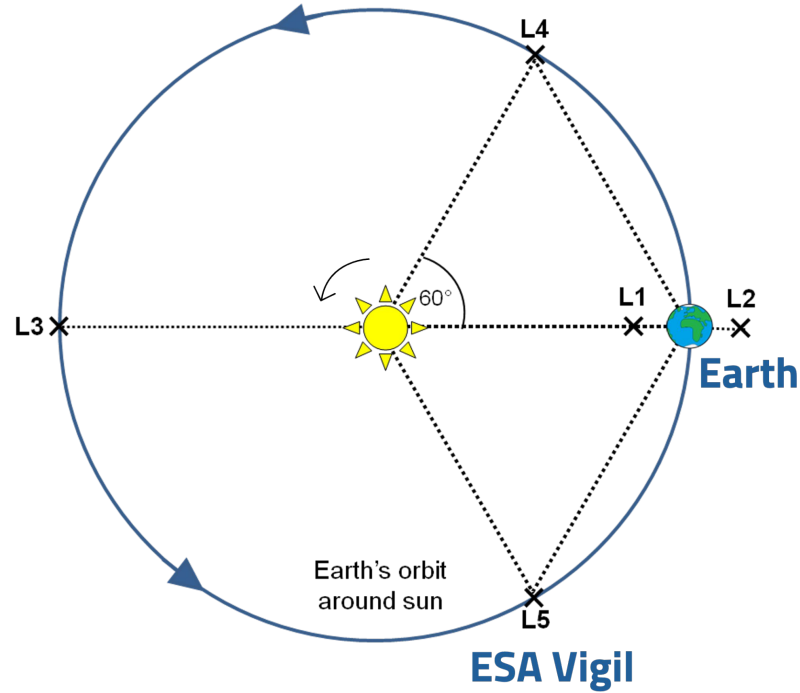


Overview

- planned for launch in 2031 on Ariane 6
 - operational after 21 months
 - expected lifetime 5 years
 - mass 2500 kg
-
- designed to operate through the worst space weather events
 - transferring data 24/7 from deep space without interruptions
 - Space Safety Programme (S2P)



Lagrangian points

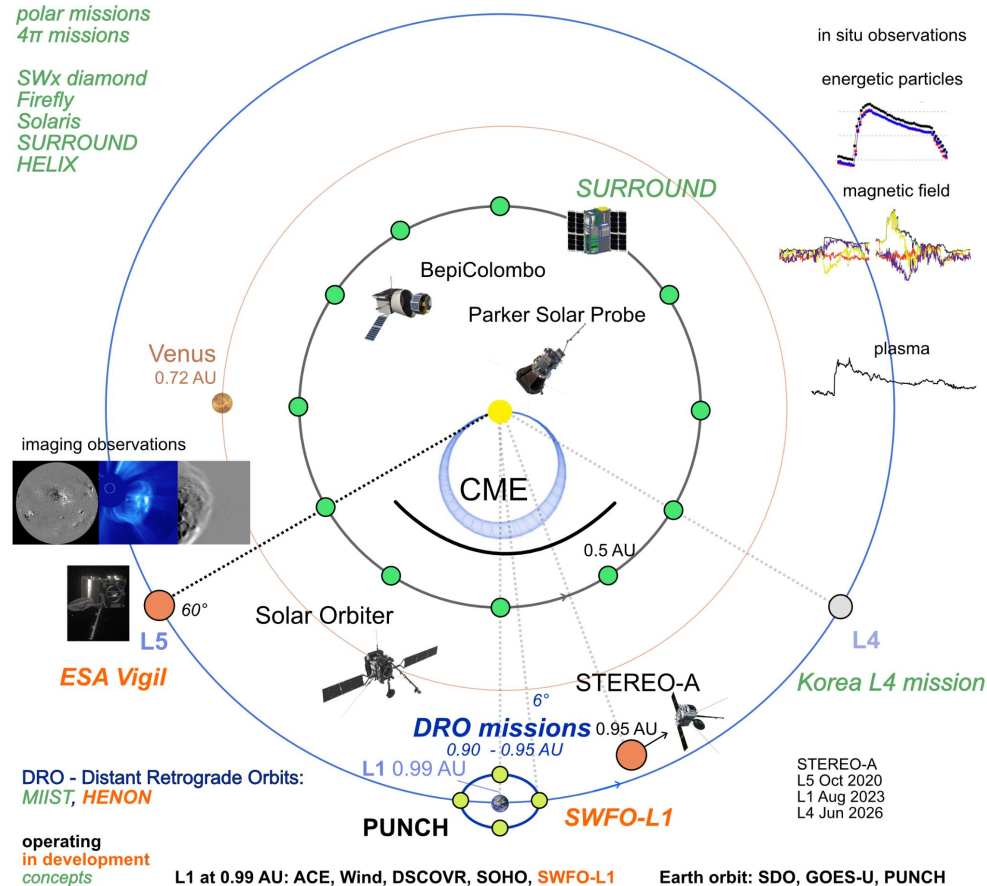


Instruments

	Instrument	Country (organization)
Remote sensing	<i>Compact Coronagraph</i>	USA (NOAA)
	<i>Heliospheric Imager</i>	Italy, Belgium
	<i>Photospheric Magnetographic Imager</i>	Germany
	<i>Extreme Ultraviolet (EUV) Imager 'JEDI'</i>	USA (NASA)
In-situ	<i>Plasma Analyser</i>	United Kingdom
	<i>Magnetometer</i>	United Kingdom, Austria

Vigil	SOHO	SDO	SoLO	STEREO	GOES	SWFO-1	Punch	WIND	Juice	ACE	DSCOVR
PMI	MDI	HMI	PHI								
NIO	EIT	AIA	EUI	SECCHI EUV	SUVI						
CCOR	LASCO				CCOR	CCOR					
HI			HI	SECCHI			HI				
MAG			MAG					MFI	IMAG	MAG	MAG
PLA			SWA					SWE		SWEPAM	FC

Space Weather and Heliophysics missions



Missions Timeline

		1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
SOHO/ EIT, LASCO, MDI	imaging																					
SDO/AIA,HMI	imaging																					
STEREO A	imaging, in-situ																					
STEREO B	imaging, in-situ																					
Solar Orbiter	imaging, in-situ																					
GOES16/SUVI	imaging, in-situ																					
GOES17/SUVI	imaging, in-situ																					
GOES18/SUVI	imaging, in-situ																					
Wind	in-situ																					
DSCOVR	in-situ																					
ACE	in-situ																					

Missions Timeline

		2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
SOHO/ EIT, LASCO, MDI	imaging																					
SDO/AIA,HMI	imaging																					
STEREO A	imaging, in-situ								?													
STEREO B	imaging, in-situ																					
Solar Orbiter	imaging, in-situ																					
GOES16/SUVI	imaging, in-situ																					
GOES17/SUVI	imaging, in-situ																					
GOES18/SUVI	imaging, in-situ																					
GOES-U/SUVI	imaging, in-situ																					
Wind	in-situ								?													
DSCOVR	in-situ								?													
ACE	in-situ																					
SWFO-1	imaging, in-situ																					
Punch	imaging									?												

Space Weather Events

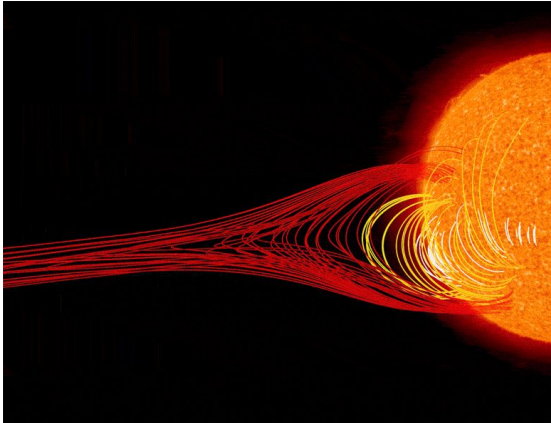
SWE drivers	SWE events	Consequences	Affected area	Scale*
EM waves	Flares, Radio bursts	SID, Radio blackout	Illuminated atmosphere	R1–R5
Plasma	CME, CIR	Geomagnetic storms	High latitudes	G1–G5
Particles	SEP	Radiation, PCA, GLE	High altitudes and latitudes	S1–S5

*as defined at: <https://www.swpc.noaa.gov/noaa-scales-explanation>

CME (coronal mass ejection)

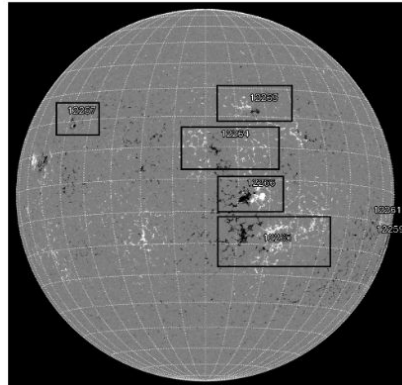
- large expulsions of plasma and magnetic field from the Sun's corona
- arrival time: 18-96h

Magnetic reconnection



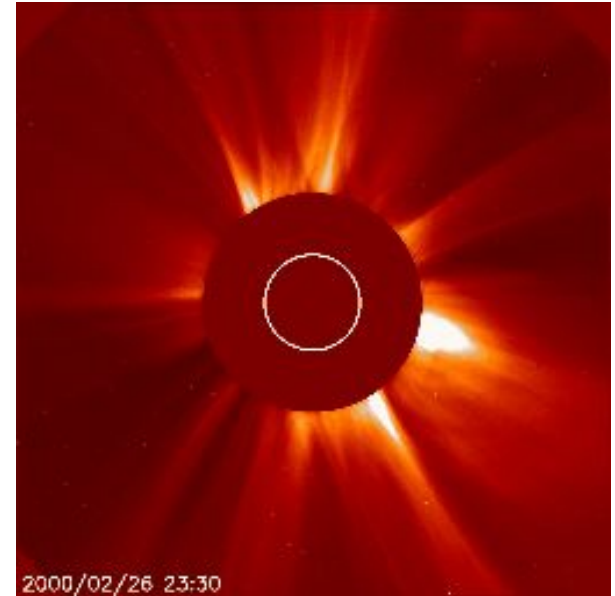
<https://scitechdaily.com/images/Simulation-of-the-Evolution-of-a-Stealth-Coronal-Mass-Ejection.gif>

Active regions, SDO/HMI



<https://www.researchgate.net/publication/354679850/figure/fig1/AS:11431281415812831@1746045446286/Full-disk-magnetogram-with-labeled-active-regions-on-20-January-2015.tif>

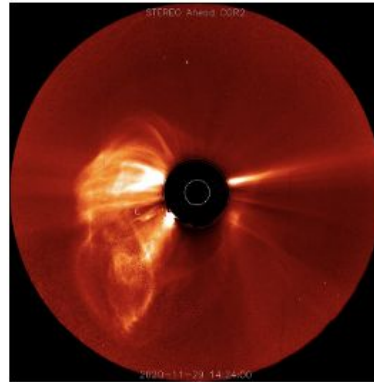
SOHO/LASCO



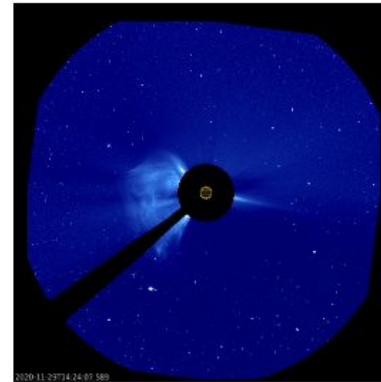
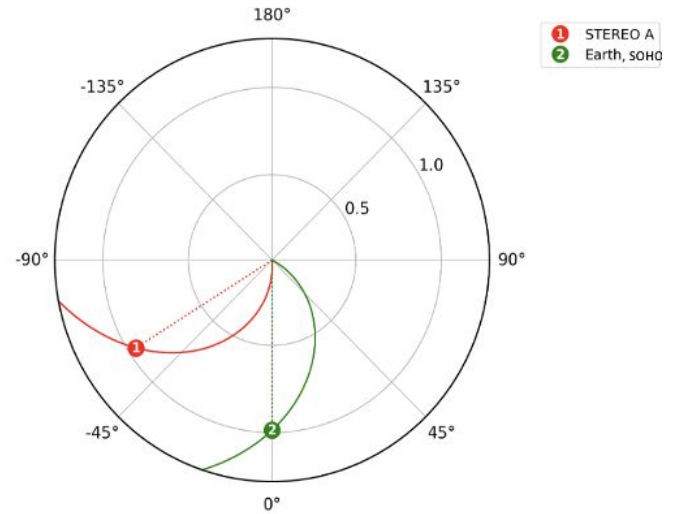
<https://www.spaceweatherlive.com/images/help/CME.gif>

2020-11-29 14:30 (UTC)

L5 & L1 comparison

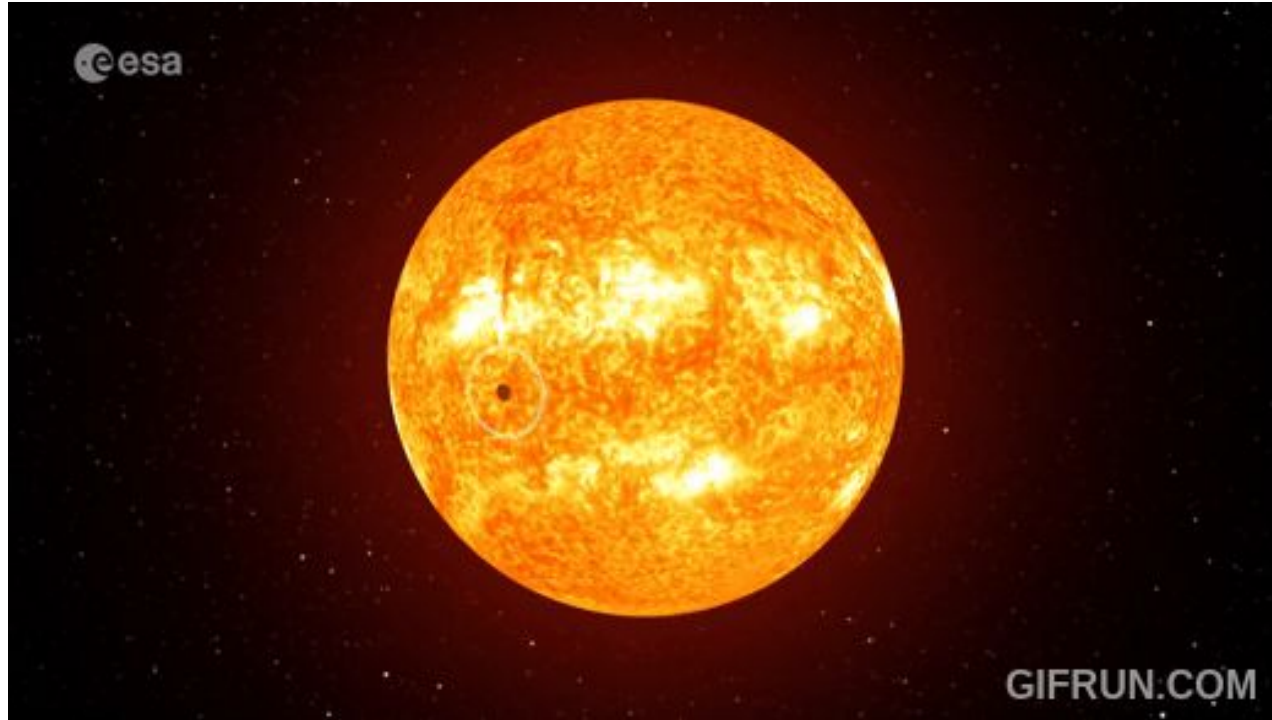


1 STEREO-A COR2



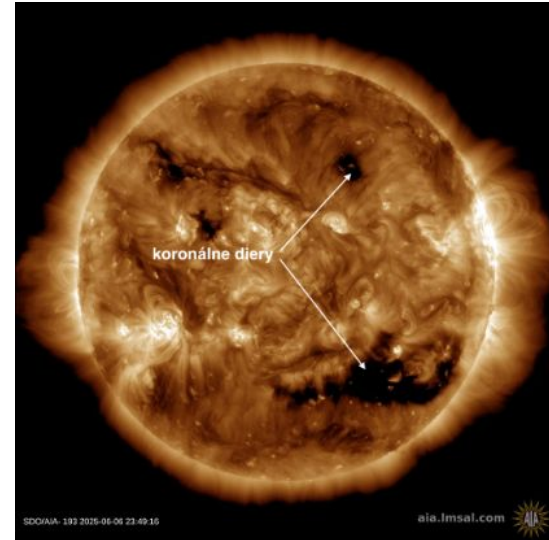
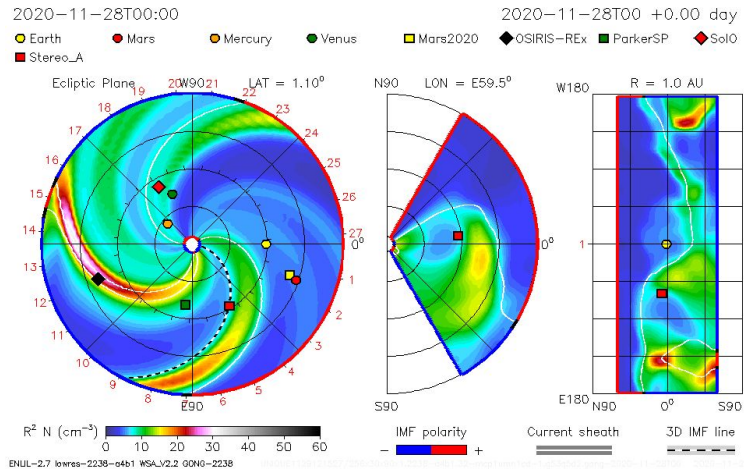
2 SOHO LASCO C2

View from L5

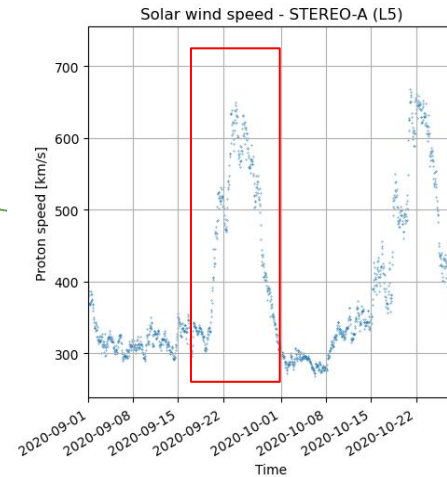
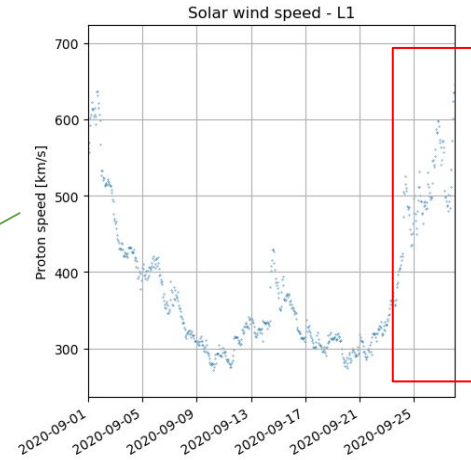
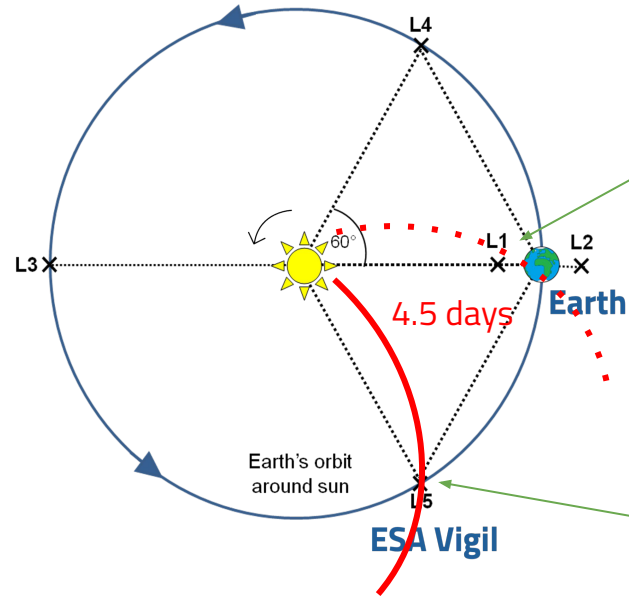


SIR/CIR

SIRs/CIRs form when fast solar wind streams originating from coronal holes interact with slower solar wind ahead of them, creating compressed regions of plasma and magnetic fields

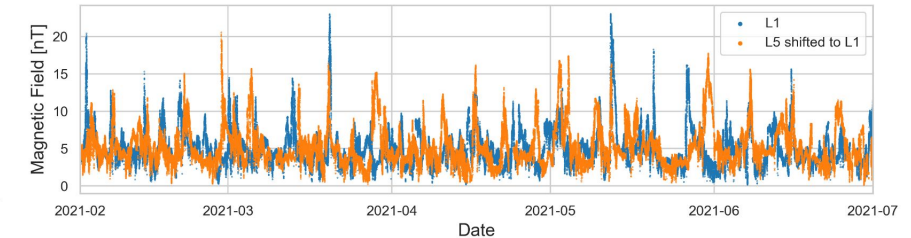
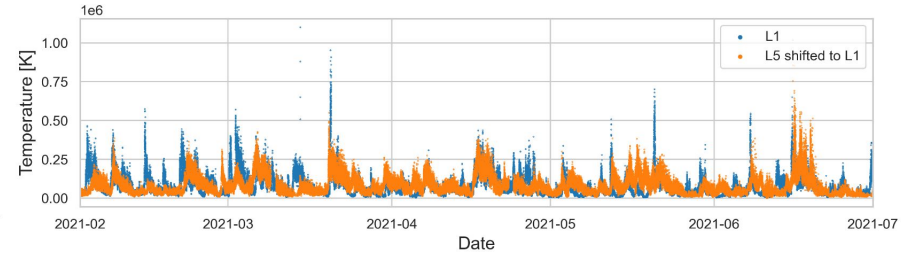
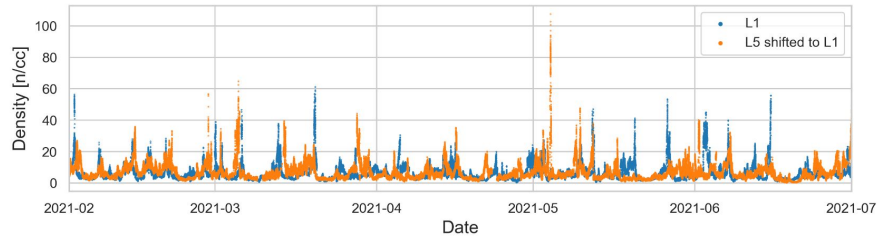
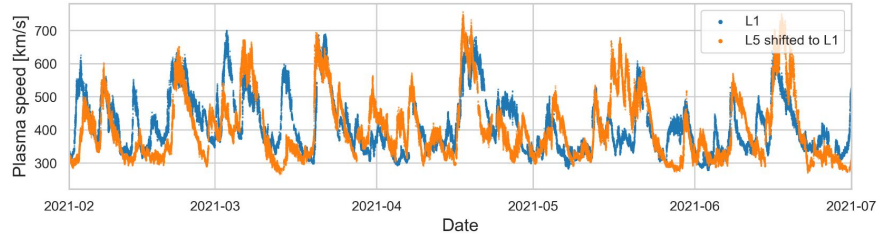


SIR/CIR



Solar wind at L5 and L1

Comparison of solar wind parameters between L1 and L5 shifted to L1



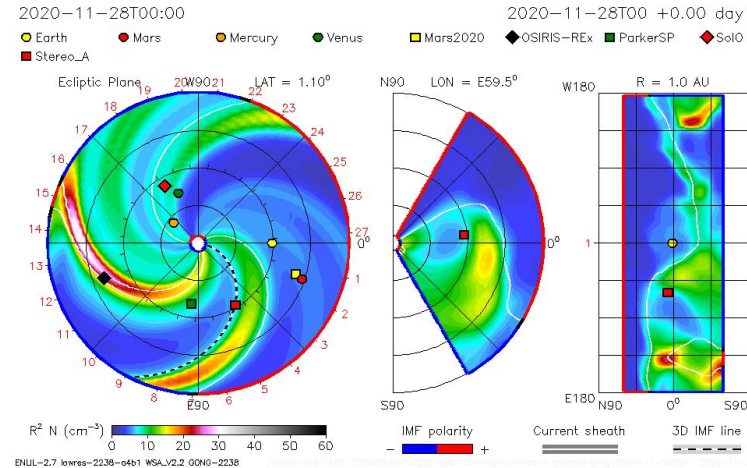
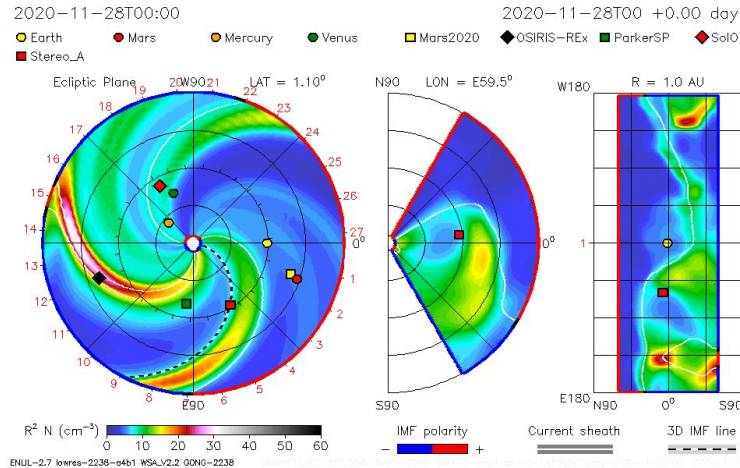
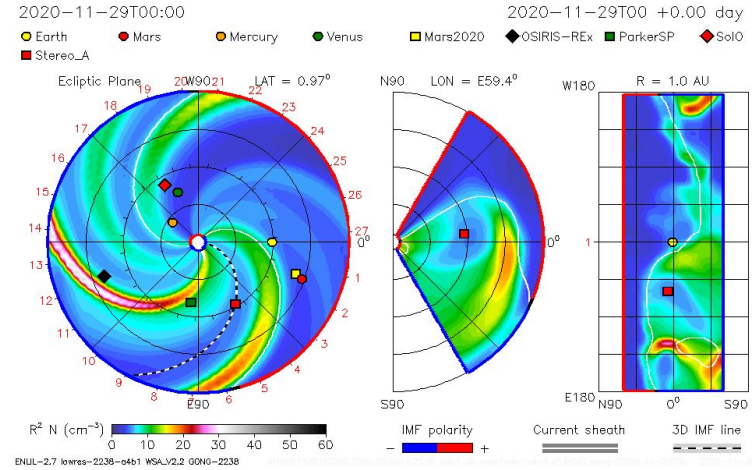
start date	end date	L1	L5
24.01.2008	24.09.2008	STEREO-A	STEREO-B
01.03.2009	26.06.2010	STEREO-A	Earth
19.12.2008	21.09.2010	Earth	STEREO-B
17.09.2013	09.03.2014	STEREO-B	STEREO-A
28.03.2020	27.07.2021	Earth	STEREO-A

$$\Delta t = \frac{\Delta \text{lon}}{\Omega_{\text{Sun}}} + \frac{r_{\text{L1}} - r_{\text{L5}}}{v_{\text{sw}}}$$

Physics-Based Model

WSA-ENLIL

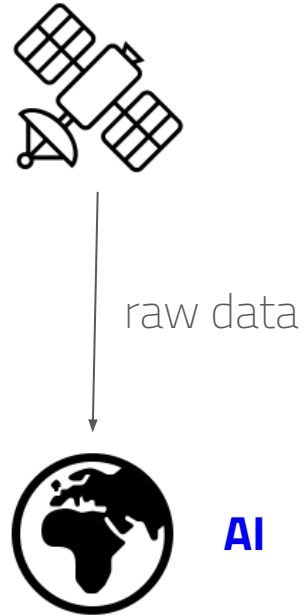
CME start 2020-11-29 13:24



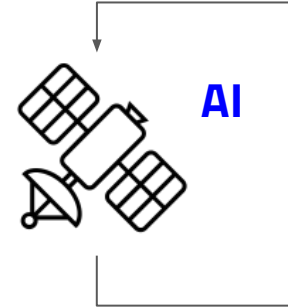
Vigil & AI



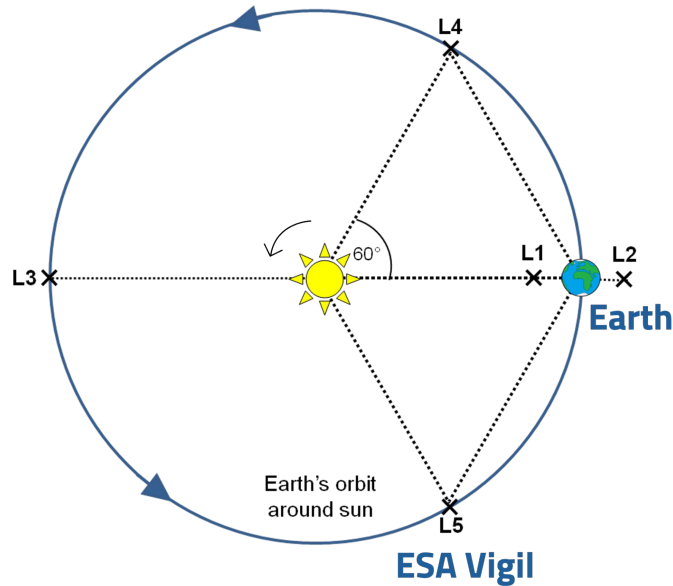
AI for satellites



Edge AI



Vigil & Space Weather Monitoring



Key advantages of Vigil:

- Ability to observe parts of the solar surface before they become visible from Earth
- Observation of CMEs from the side as they propagate from the Sun toward Earth
- Measurements of the IMF and solar wind features rotating toward Earth

Vigil & Space Weather Monitoring

- *For the VIGIL mission, ESA has developed a dedicated “computational memory” space grade hardware and an Artificial Intelligence ensemble system tool to automatically classify solar flares.*
- *With current instruments, we often wait five or six hours for downlinks. Vigil’s low-latency system allows for rapid data transfer, which means we can cut forecast errors from 24 hours down to two.*
- *We expect a Vigil II, III, and beyond. Multiple points of observation will help us build 3D models of solar activity—making our forecasts sharper, faster, and more accurate.*

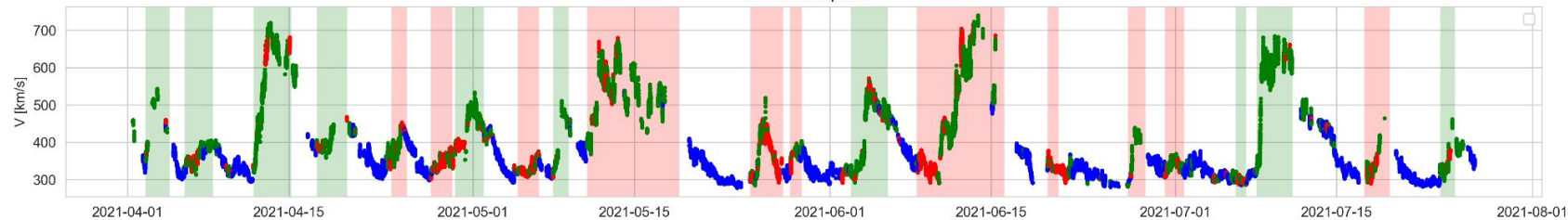
Automatic (geoeffective) CIR detection

CIRs catalog

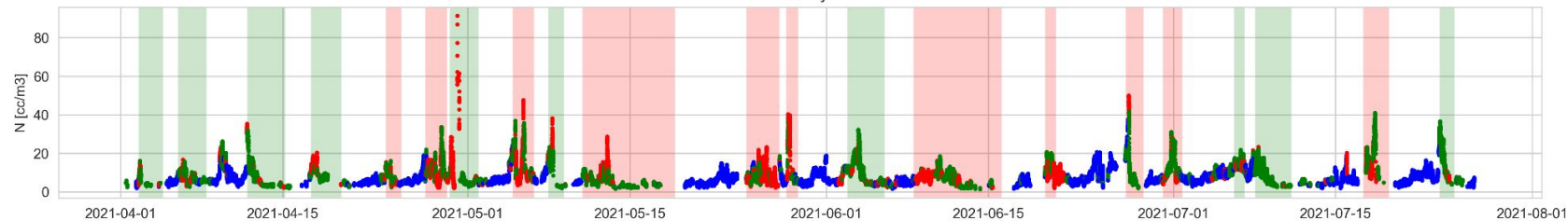
- 28.3.2020 – 27.7.2021
- 59 CIRs

ID	Start Time	End Time	Interval End Criterion	Type	L1 Occurrence	$\Delta t_{\max}(P_t)$ L5–L1	Coronal Hole No. (SWPC Synoptic Maps)
1	2020-03-24 16:19	2020-03-28 12:09	$V_p < 400$	A	0		24
2	2020-04-02 05:43	2020-04-02 11:50	$P_t < 20$	B	CIR	-10.5H	25
3	2020-04-05 09:55	2020-04-06 10:12	$P_t < 20$	B	CIR	-13.0H	27
4	2020-04-09 00:27	2020-04-09 19:20	$P_t < 20$	B	0		26
5	2020-04-15 07:50	2020-04-15 23:34	$V_p < 400$	A	0		30
6	2020-04-19 21:26	2020-04-25 17:35	$P_t < 20$	M	0		31, 32
7	2020-04-29 11:55	2020-04-30 05:41	$P_t < 20$	B	CIR	-7.3H	34, 35
8	2020-05-03 06:08	2020-05-04 16:57	$0.8 \max(V_p)$	A	0		36
9	2020-05-09 07:41	2020-05-09 20:08	$P_t < 20$	B	CIR	+3.2H	37
10	2020-05-12 21:22	2020-05-15 19:33	$V_p < 400$	M	CIR	-11.1H	38, 39
11	2020-05-16 16:40	2020-05-18 09:37	$0.8 \max(V_p)$	A	CIR	+14.7H	40
12	2020-05-23 18:40	2020-05-23 20:57	$P_t < 20$	B	0		41
13	2020-05-25 14:49	2020-05-27 13:53	$V_p < 400$	M	0		42, 43, 44, 45
14	2020-06-01 17:34	2020-06-02 06:57	$P_t < 20$	B	CIR	+1.0H	46

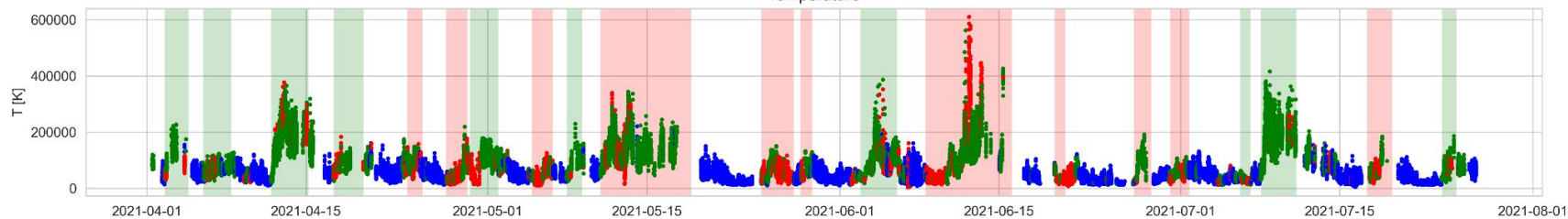
Solar wind speed



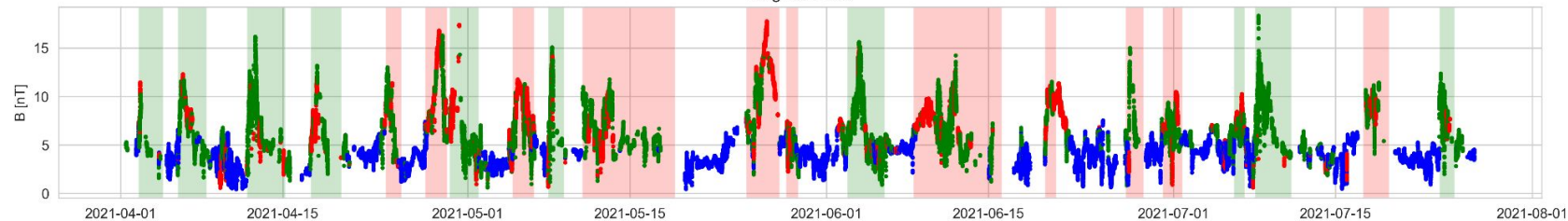
Density



Temperature

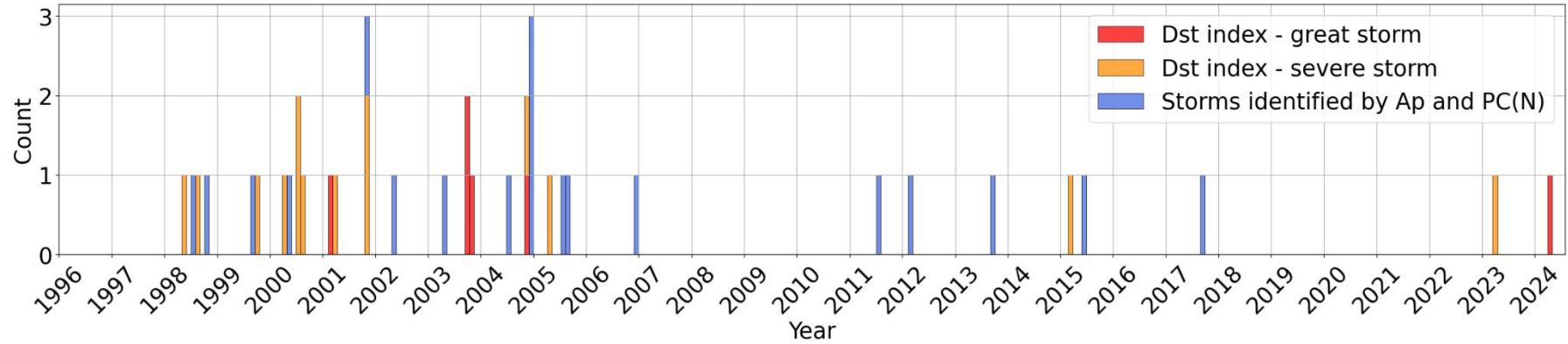


Magnetic field



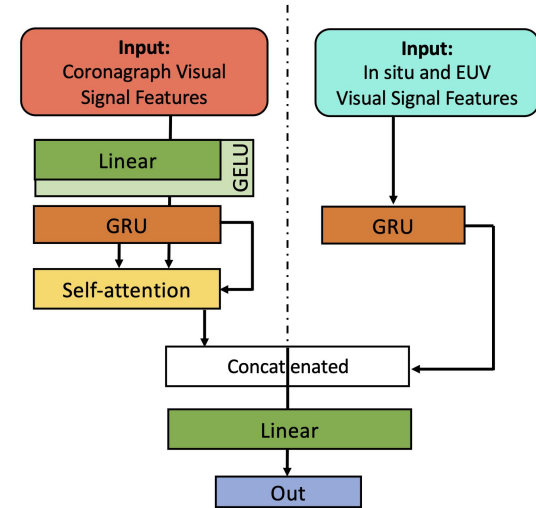
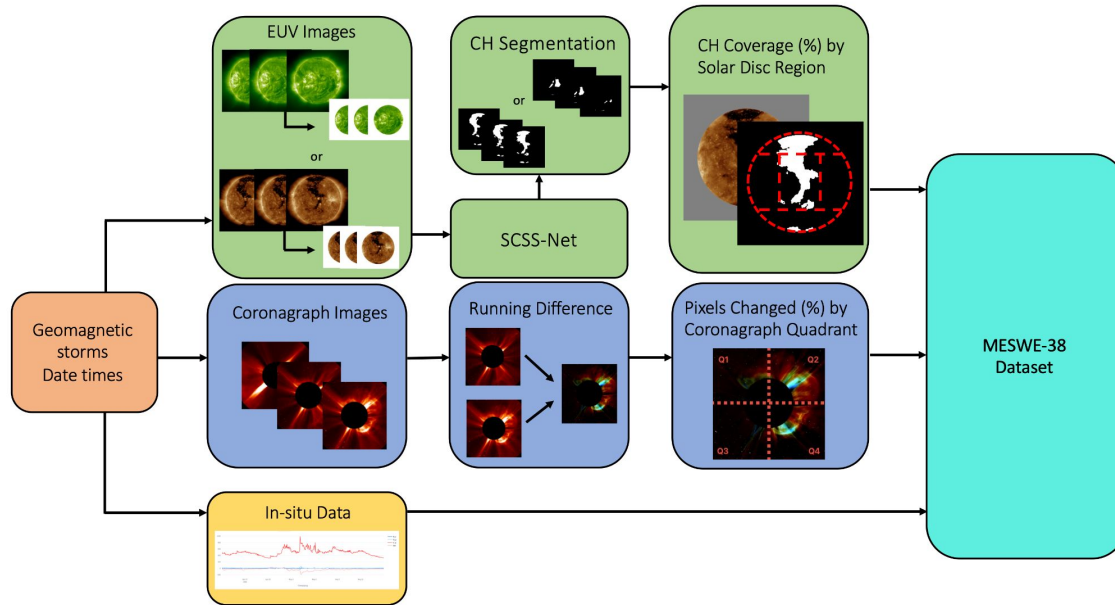
Dst prediction

most extreme space weather events over last 30 years



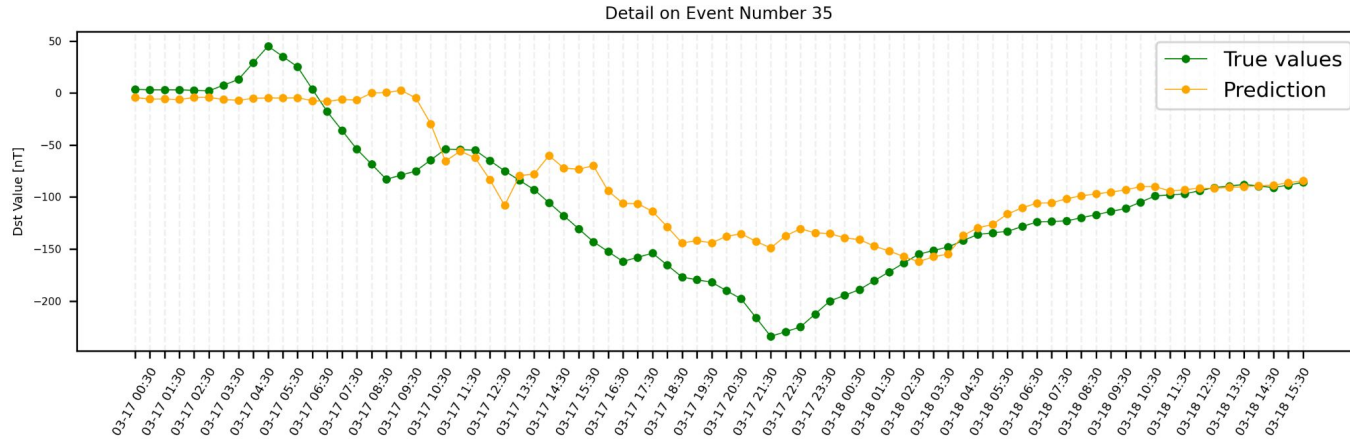
Parameter	Severe storms	Great storms
Dst index	$-200 \text{ nT} > \text{Dst} > -350 \text{ nT}$	$-350 \text{ nT} \geq \text{Dst}$
Ap index	$179 \leq \text{Ap} \leq 300$	$300 < \text{Ap}$
PCN index	$10 \text{ mV/m} < \text{PCN} \leq 15 \text{ mV/m}$	$15 \text{ mV/m} < \text{PCN}$

Dst prediction

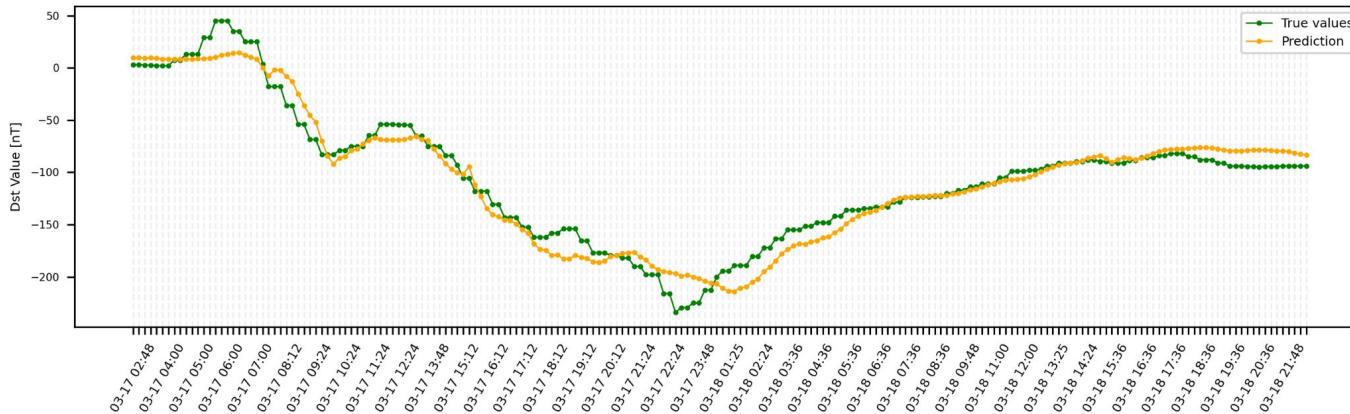


Dst prediction

community
benchmark



our model



Thank you for your attention